Basic Concepts in Logic

2.1. Arguments, Premises and Conclusions

Is reasoning measurable?

- Philosophy is not a mere speculation.

- It depends on intense analysis which involves rigorous use of reasoning.

- Without logical reasoning philosophy becomes a matter of opinion.

- There are objective and measurable rules of reasoning.
- Logic is the systematic study of these rules.

So, what is reasoning?
And what is the difference between reasoning and argument?

It’s the process of drawing, subtracting, deducing, extracting or inferring something out of something which is already available in some form.

Example 1.



Example 2.

B+B= 4

B+T= 5

B+C= 6

B+C+T+T=?

Example 3:
*NASA received three messages in a strange language from distant planet. The scientists studied the messages and found that "**Necor Buldon Slock**" means "**Rocket Danger Explosion**" and "**Buldon Gimilzor Gondor**" means "**Rocket Spaceship Fire**" and "**Edwan Mynor Necor**" means "**Bad Gas Explosion**". What does "**Slock**" mean?*

- Reasoning/Inference can be used interchangeably with argument.

- An inference, in the technical sense of the term, is the reasoning process expressed by an argument.

- Inference/reasoning refers to the mind act or the mental process while argument is its verbal expression.

The ordinary definition of argument: altercation, dispute, confrontation, noisy controversy, bickering, quarrel, intense and fierce verbal fight or exchange, heated debate which involves shouting.

- **Argument** is a group of statements one or more of which provide supporting evidence to another statement.
- It is an inferential relationship between statements.
- The statement(s) which provide support are called **premise(s)**
- the statement to be supported is known as **conclusion**.
- Premises set forth evidence in defense of the conclusion.
- So, argument is a process of drawing a conclusion from a premise or set of premises.

•**Statements** are the building blocks of arguments.

•**Statement** is any declarative sentence with truth value.

•**Proposition** vs. **Statement**

•In the technical sense, the former is the meaning or information content of a statement whereas the latter is its verbal expression.

Example:

Grass is Green.

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- **Questions** (What shall we eat for lunch?)
- **Commands** (Close the window!)
- **Propositions** (Let’s go to a movie tonight.)
- **Suggestions** (I suggest you wear sunglasses.)
- **Exclamations** (Alas! Oh my God!)

- What comes next is how to distinguish premise(s) from conclusions in an argument.

- *premise and conclusion indicator words:*
  - **Premise indicators***: for, since, as, owing to, in as much as, …*
  - **Conclusion indicators***: so, thus, therefore, hence, accordingly,…*
  - *Process of elimination*
  - *Adding one's own indicator*

Addressing the question

- What single statement is claimed (implicitly) to follow from the others?
- What is the arguer trying to prove?
- What is the main point of the passage?

Example 1

***X**: Abortion is wrong.*
***Y**: No it isn't!*
***X**: Yes it is!*
***Y**: No it is not!*
***X**: You fool! What do you know anyway?*
***Y**: Well, at least I know more than you do. You idiot!*

Example 2

Since murder is wrong and abortion is a form of murder; it follows that, abortion is wrong.

Example 3

In this paragraph we shall discuss the benefits of sleep. It really does matter if you get enough sleep. We need sleep to think clearly, react quickly and create memories. Studies show that people who are taught mentally challenging tasks do better after a good night's sleep. Other research suggests that sleep is needed for creative problem solving. By the way when we discussed the benefits of sleep those students who frequently sit at the back in the class were asleep.

Example 4

Society values white lives more than black lives. This is clear from the fact that killers of whites are much more likely to be sentenced to death than killers of blacks. Of the 1788 people currently on death row 1713 were convicted of killing a white person. Yet blacks are six times more likely to be murder victims than whites are. In Florida, no one has ever been executed for murdering a black person but dozens have been executed for murdering white people. This is the comparison in short.

2.2. RECOGNIZING ARGUMENTS

- A passage contains an argument if it purports to prove something.
  - Two conditions must be fulfilled for a passage to purport to prove something:

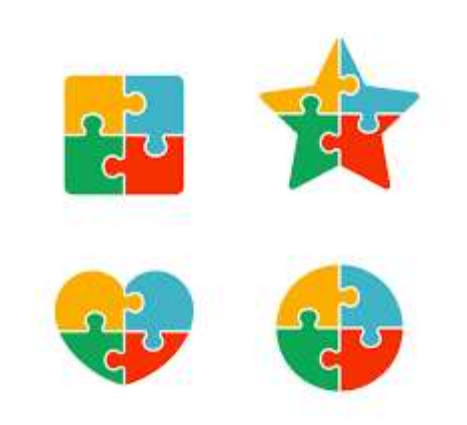
1. *At least one of the statements must claim to present evidence or reasons.*
2. *There must be a claim that the alleged evidence or reasons supports or implies something- that is, a claim that something follows from the alleged evidence.*

- The first condition is said to be a **factual claim** whereas the second condition is called **inferential claim**.
- Factual claim is about the factuality of statements.
- Inferential claim is about the conclusion claimed to follow from premises
- Logic is chiefly interested in inferential claim.

What do you think people mean when they say ‘this is logical or illogical’?

Example 1

All Zoodles are Zonkers.
Helen is a Zoodle.
Therefore, Helen is a Zonker.





➢We can identify a passage with an argument in the following **three** ways. These are:

1.Premise and conclusion indicators

2.Inference

3.Typical Non-arguments

i. Warning
ii. Piece of Advice
iii. Statements of Opinion or Belief
iv. Loosely associated statements
v. Report
vi. Expository passage
vii. Illustration
viii. Conditional statements
ix. Explanation

Argument Vs. Explanation

Argument

Premise(s) Accepted facts

⬇ claimed to prove

Conclusion

- There is an inferential link between *premises* and *conclusion*
- It proves *that* something is the case
- It is used when there is doubt about the truth of the conclusion.

Explanation

Explanans

⬇ - claimed to shed light on

Explanandum - Accepted fact

- There is a rational link between *explanans* and *explanandum*
- It shows *why* something is the case. It does not purport to prove anything.
- It is used to shed light on an already known or accepted fact.

2.3. DEDUCTION AND INDUCTION

- Argument or reasoning is broadly classified into **two**.
  - This classification depends on the inferential claim and the nature of the support the premises provide in defense of the conclusion.
  - The strength of the inferential claim
  - If the conclusion is claimed to follow with strict certainty or necessity, the argument is said to be deductive, but if it is claimed to follow only probably, the argument is inductive.

Deduction Vs. Induction

- Is an argument in which the premises are claimed to support the conclusion in such a way that if they are assumed true, it is *impossible* for the conclusion to be false. If the premises are true, then the conclusion *must* be true.
- The conclusion is claimed to follow *necessarily* from the premises.
- The conclusion is already contained in the premises.
- It is knowing after fact
- It goes from familiar to familiar
- Truth is guaranteed!
- Necessary Reasoning

- Is an argument in which the premises are claimed to support the conclusion in such a way that if they are assumed true, then based on that assumption it is *improbable* that the conclusion is false. If the premises are true it is *probable* that the conclusion is true.
- The conclusion is claimed to follow only *probably* from the premises.
- The content of the conclusion is in some way intended to 'go beyond' the content of the premises.
- It is knowing before fact
- It goes from relatively familiar to less familiar or proceeds from presumably known to the presumably unknown.
- Estimation of truth.
- Probabilistic reasoning

There are **three** ways by which we distinguish deduction from induction or vice versa.

1. the occurrence of special indicator words
2. the actual strength of the inferential link between premises and conclusion
3. the character or form or style of argumentation the arguer uses

Deductive Vs. Inductive Form/Style

- Mathematical reasoning
- Argument based on Definition
- Categorical Syllogism
- Hypothetical Syllogism
- Disjunctive Syllogism

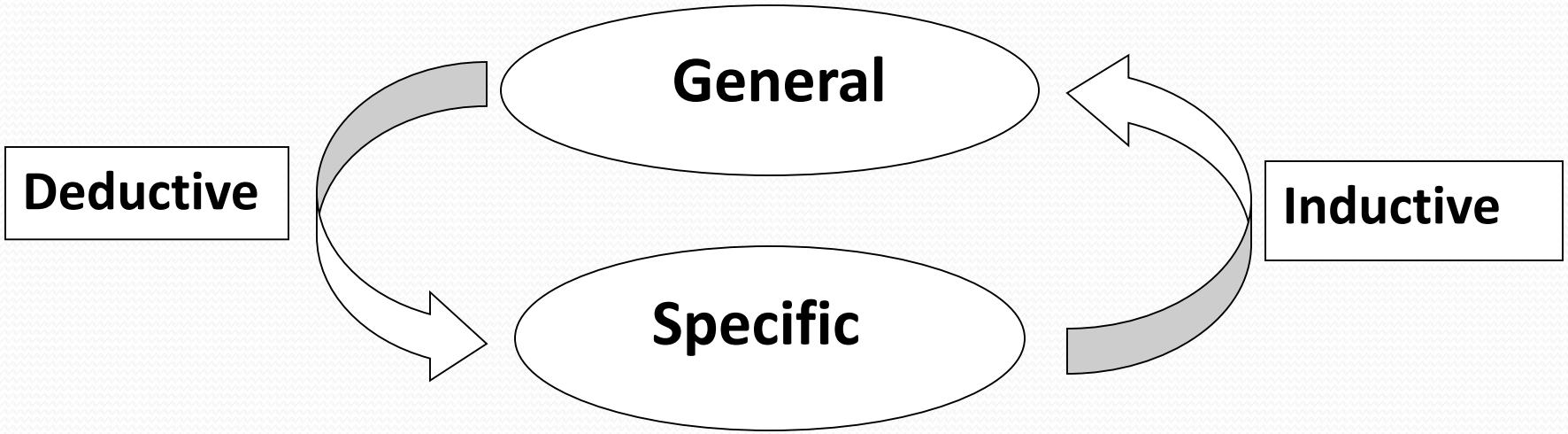
- Prediction
- Argument from analogy
- Inductive generalization
- Argument from authority
- Argument based on signs
- Causal inference

Scientific Arguments

- Arguments aimed at the ***discovery*** of a law of nature are usually considered inductive.

- Arguments aimed at the ***application*** of known laws to specific circumstances are best interpreted as deductive.

Traditional Definition of Deduction and Induction



Examples

1. *All Africans are rich. All rich people are hard workers. Therefore, all Africans are hard workers.*
2. *Bobby is a dog. Bobby has a tail. Therefore, Bobby's tail is the tail of a dog.*
3. 3 is a prime number.*5 is a prime number.7 is a prime number. Therefore, all odd numbers between 2 and 8 are prime numbers.*
4. *All diamonds found previously were gray. Therefore, the next diamond to be found will be gray.*
5. *The members of the Williams family are Susan, Nathan and Alexander. Susan wears glasses. Nathan wears glasses. Alexander wears glasses. Therefore, all members of the Williams family wear glasses.*

2.4. EVALUATING ARGUMENTS: VALIDITY,TRUTH, SOUNDNESS AND COGENCY

- In evaluating arguments, we need to focus on two particular aspects of arguments.
  - Quality of the reasoning
    [Is the argument *logically* good?]
  - The truth of the premises.
    [Is the argument *factually* good?]
    [Are the statements *factually* correct?]

Valid
Invalid
} Inferential claim

Sound
Unsound
} Factual claim

} Deductive

Strong,
Weak,
} Inferential claim

Cogent,
Uncogent
} Factual Claim

} Inductive

Validity

- A **valid** argument is one in which the premises support the conclusion completely.
- A valid argument is one such that it is impossible for its conclusion to be false while its premises are true.

- An **invalid** argument has this essential feature: It is possible for its conclusion to be false while its premises are true. An argument with actually true premise and actually false conclusion is invalid.

Compare and contrast the following examples:

1. All human beings are mortal. Socrates is a human being. Therefore, Socrates is mortal.
2. All human beings are mortal. Socrates is mortal. Therefore, Socrates is a human being.
3. Since this chalk is red. So, the chalk is colored.
4. Since this chalk is colored; it follows that it is red.

- Notice that an argument can have one or more false premises and still be valid.
- Notice that an argument can have true premises but not valid.
- Notice that an argument can have true premises and true conclusion but not valid.
- Notice that an argument can have false premises and false conclusion but not invalid.

premise	Conclusion	validity
True	True	?
True	False	Invalid
False	True	?
false	false	?

Sound and Unsound Reasoning

❖A deductively **sound** argument has two essential features: It is valid and all its premises are true.

Notice that a deductively sound argument cannot have a false conclusion. Because a sound argument is valid and has only true premises, it must have a true conclusion.

❖A deductively **unsound** argument falls into one of the following three categories:

- It is valid but has at least one false premise.
- It is invalid but all its premises are true.
- It is invalid and has at least one false premise.

In other words, a deductively unsound argument is one that either is invalid or has at least one false premise.

Sound = valid + All true premises

Strength, Weakness, Truth, and falsity

- An inductive argument is one in which the premises are claimed to support the conclusion in such a way that if they are assumed true, then based on that assumption it is only probable that the conclusion is true. If the premises do in fact support the conclusion in this way, the argument is said to be **strong**.

- Conversely, a **weak** inductive argument is an inductive argument such that if the premises are assumed true, then based on that assumption it is not probable that the conclusion is true.

- Notice that any inductive argument having *actually* true premises and a probably false conclusion is always weak.

premise	conclusion	strength
True	Probably true	?
True	Probably False	Weak
False	Probably True	?
false	Probably false	?

A **cogent** argument is an inductive argument that is strong and has all true premises; if either condition is missing the argument is **uncogent**. In other words, it is equivalent of sound deductive argument. Thus, an uncogent argument is an inductive argument that is weak, has one or more false premises, or both.

Cogent = **Strong** + **All true premises**

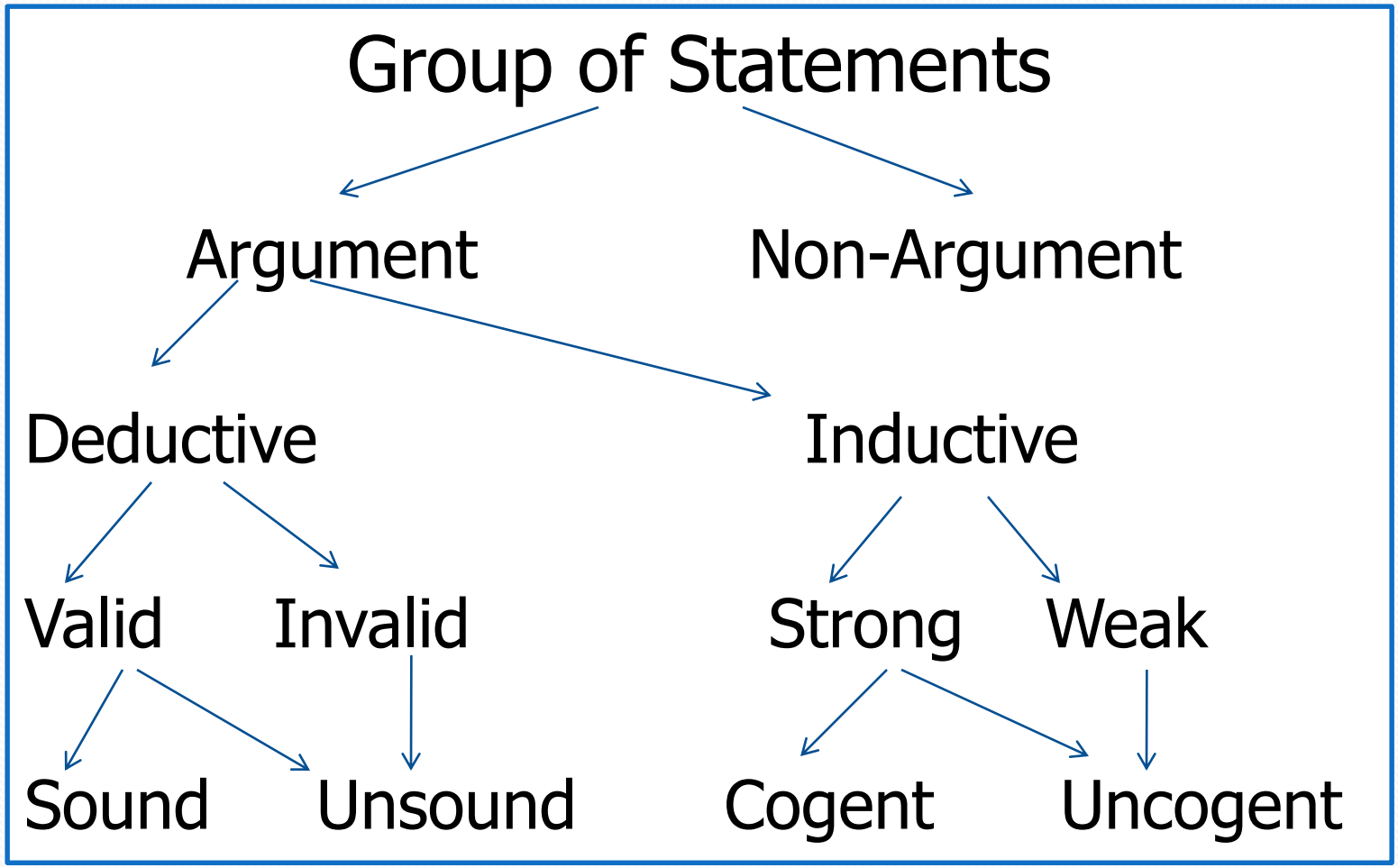
Notice that inductive reasoning in general and cogency in particular require **"uniformity of nature"** and presenting **"total evidence."**

In summary, for both deductive and inductive arguments two separate questions need to be answered:

Do the premises support the conclusion?

Are all the premises true?

The first question concerns the truth of an argument's inferential claim; that is, the correctness of an arguments reasoning. Its answer determines whether a deductive argument is valid or invalid and whether an inductive argument is strong or weak. The second question concerns the truth of an argument's factual claim. Assuming an argument is valid or strong, the answer to the second question determines whether a deductive argument is sound or unsound and whether an inductive argument is cogent or uncogent.


Group of Statements
Argument
Non-Argument
Deductive
Inductive
Valid
Invalid
Strong
Weak
Sound
Unsound
Cogent
Uncogent